Course Code:	Course Title	Credit
CSC701	Machine Learning	3

Prerequisite: Engineering Mathematics, Data Structures, Algorithms	
Course Objectives:	
1	To introduce the basic concepts and techniques of Machine Learning.
2	To acquire in depth understanding of various supervised and unsupervised algorithms
3	To be able to apply various ensemble techniques for combining ML models.
4	To demonstrate dimensionality reduction techniques.
Course Outcomes:	
1	To acquire fundamental knowledge of developing machine learning models.
2	To select, apply and evaluate an appropriate machine learning model for the given
3	To demonstrate ensemble techniques to combine predictions from different models.
4	To demonstrate the dimensionality reduction techniques.

Module		Content	Hrs
1		Introduction to Machine Learning	04
	1.1	Machine Learning, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing a Machine Learning Application.	
	1.2	Training Error, Generalization error, Overfitting, Underfitting, Bias-Variance trade-off.	
2		Learning with Regression and Trees	09
	2.1	Learning with Regression: Linear Regression, Multivariate Linear Regression, Logistic Regression.	
	2.2	Learning with Trees: Decision Trees, Constructing Decision Trees using Gini Index (Regression), Classification and Regression Trees (CART)	
	2.3	Performance Metrics: Confusion Matrix, [Kappa Statistics], Sensitivity, Specificity, Precision, Recall, F-measure, ROC curve	
3		Ensemble Learning	06
	3.1	Understanding Ensembles, K-fold cross validation, Boosting, Stumping, XGBoost	
	3.2	Bagging, Subagging, Random Forest, Comparison with Boosting, Different ways to combine classifiers	
4		Learning with Classification	08
	4.1	Support Vector Machine Constrained Optimization, Optimal decision boundary, Margins and support vectors, SVM as constrained optimization problem, Quadratic Programming, SVM for linear and nonlinear classification, Basics of	

		Kernel trick.	
	4.2	Support Vector Regression, Multiclass Classification	
5		Learning with Clustering	07
	5.1	Introduction to clustering with overview of distance metrics and major clustering approaches.	
	5.2	Graph Based Clustering: Clustering with minimal spanning tree Model based Clustering: Expectation Maximization Algorithm, Density Based Clustering: DBSCAN	
6		Dimensionality Reduction	05
	6.1	Dimensionality Reduction Techniques, Principal Component Analysis, Linear Discriminant Analysis, Singular Valued Decomposition.	
		Total	39

Textbooks:	
1	Peter Harrington, “Machine Learning n Action”, DreamTech Press
2	Ethem Alpaydın, “Introduction to Machine Learning”, MIT Press
3	Tom M. Mitchell, “Machine Learning” McGraw Hill
4	Stephen Marsland, “Machine Learning An Algorithmic Perspective”, CRC Press
References:	
1	Han Kamber, ―Data Mining Concepts and Techniques‖, Morgan Kaufmann Publishers
2	Margaret. H. Dunham, ―Data Mining Introductory and Advanced Topics, Pearson Education
3	Kevin P. Murphy , Machine Learning ― A Probabilistic Perspective‖
4	Samir Roy and Chakraborty, ―Introduction to soft computing‖, Pearson Edition.
5	Richard Duda, Peter Hart, David G. Stork, “Pattern Classification”, Second Edition, Wiley Publications.
Assessment:	
Internal Assessment:	
Assessment consists of two class tests of 20 marks each. The first class test is to be conducted when approximately 40% syllabus is completed and the second class test when an additional 40% syllabus is completed. Duration of each test shall be one hour.	
End Semester Theory Examination:	
1	Question paper will comprise a total of six questions.
2	All question carries equal marks
3	Questions will be mixed in nature (for example supposed Q.2 has part (a) from module 3 then part (b) will be from any module other than module 3)
4	Only Four questions need to be solved.